Project: Worked examples
Title: Bolted joint (simplified VDI 2230)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Calculates the preload from the tightening torque, the load factor, the residual clamp, separation, yield, thread stripping and fatigue amplitude of one bolt clamping members under an axial working load. It uses a simplified VDI 2230 approach.

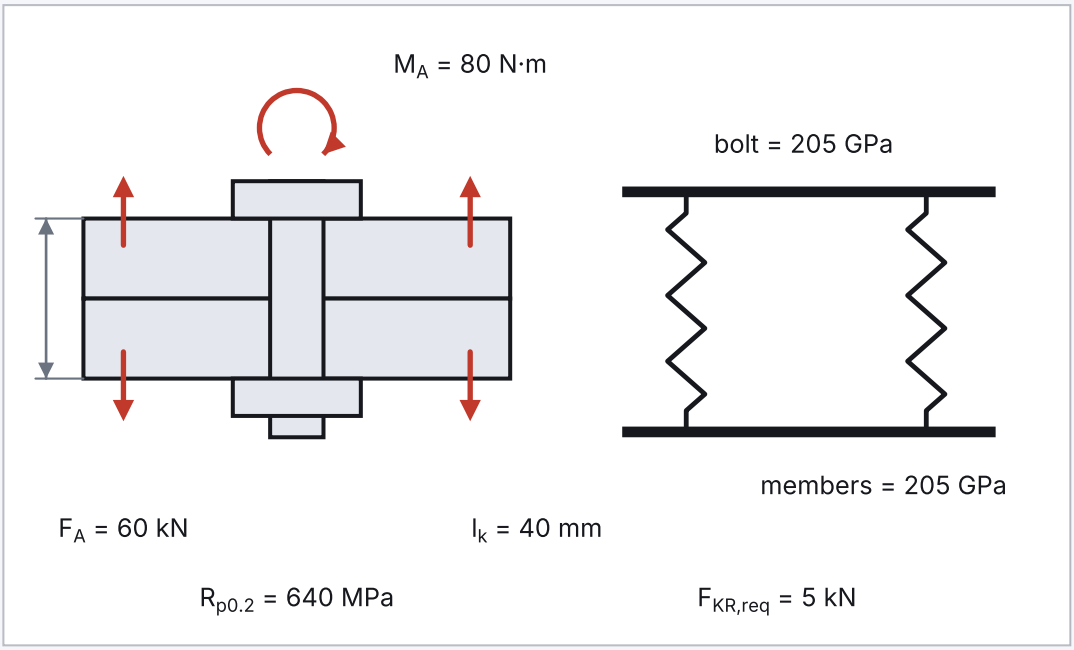



Figure 1 — A preloaded bolt clamping two members, the working load pulling them apart; beside it, bolt and members as two springs in parallel. Not to scale; the labels are the inputs as entered.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Tensile stress area | $A_s = (\pi/4)\cdot((d_2 + d_3)/2)^2$, with $d_2 = d - 0.6495\ p_{thread}$ and $d_3 = d - 1.2269\ p_{thread}$ |
| Assembly preload | $F_M = M_A/(K\ d)$ by the nut factor; $F_M = M_A/(0.16\ p_{thread} + 0.58\ d_2\ \mu_G + 0.5\ D_{Km}\ \mu_K)$ by thread and head friction; $F_M = F_V$ when measured; $F_{M,min} = F_M/\alpha_A$ after the scatter |
| Thread torsion under preload | $\tau_M = M_G/W_p$ with $M_G = F_M\ (0.16\ p_{thread} + 0.58\ d_2\ \mu_G)$ and $W_p = \pi/16\cdot d_3^3$ at the minor diameter, which is conservative against the VDI 2230 mean diameter $d_S$ |
| Bolt resilience, the head, free shank and engaged thread as springs in series | $\delta_S = (0.4\ d + 0.4\ d)/(E_S\ A_N) + l_k/(E_S\ A_s) + 0.5\ d/(E_S\ A_{d3})$ |
| Member stiffness, the clamped parts as two pressure frusta | $k = 0.5774\ \pi\ E\ d\ /\ \ln[((1.155\ t + d_w - d_h)(d_w + d_h))/((1.155\ t + d_w + d_h)(d_w - d_h))]$ |
| Load factor and embedding | $\Phi = n_{load}\ k_S/(k_S + k_P)$; $F_Z = f_{embed}\ (k_S\ k_P)/(k_S + k_P)$ |

| WHAT IT CALCULATES | RELATION |
|---|---|
| Working state | $F_{KR} = \max(F_{M,min} - (1 - \Phi)\, F_A - F_Z,\, 0)$; $F_{S,max} = \max(F_M + \Phi\, F_A,\, F_A)$; $\sigma_{red} = \sqrt{(\sigma_z^2 + 3\,(0.5\,\tau)^2)}$; $\sigma_a = (F_{S,max} - F_M)/(2\,A_s)$ |
| Thread engagement needed against stripping | $m_{min} = S\, F_{S,max}/(0.5\,\pi\, d\, \tau_B)$ |

Basis: Systematic calculation of highly stressed bolted joints — calculation steps R0–R13[1](§5.1–§5.5); Tensile stress area of the thread[2](§9.1.6); Joint stiffness (frusta); preload and torque[3](§8-5, §8-7, 11th ed.).

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| method | Preload method | friction | | spec: Assembly procedure AP-7 | any |
| d | Nominal thread diameter | 12 | mm | drawing: DRG-FLG-02 rev A | 3 … 64 mm |
| $p_{thread}$ | Thread pitch | 1.75 | mm | spec: ISO 261 coarse pitch | 0.35 … 6 mm |
| $R_{p02}$ | Bolt 0.2 % proof (yield) strength | 640 | MPa | spec: ISO 898-1 class 8.8 | 200 … 1200 MPa |
| $E_S$ | Bolt Young's modulus | 205 | GPa | spec: steel | 60 … 250 GPa |
| $\sigma_{a,limit}$ | Bolt endurance limit (stress amplitude) | 50 | MPa | assumption: Rolled-before-heat-treatment M12, VDI 2230 order of magnitude. | 10 … 200 MPa |
| $l_k$ | Clamp length | 40 | mm | drawing: DRG-FLG-02 rev A | 1 … 2000 mm |
| $d_h$ | Hole diameter | 13 | mm | drawing: DRG-FLG-02 rev A | 1 … 100 mm |
| $d_w$ | Head / washer bearing diameter | 18 | mm | spec: ISO 4762 head bearing | 2 … 200 mm |
| $E_P$ | Clamped members' Young's modulus | 205 | GPa | spec: steel | 5 … 250 GPa |
| $n_{load}$ | Load introduction factor (0 = at the clamped faces, 1 = under the head) | 0.5 | | assumption: Load introduced at mid-thickness. | 0 … 1 |
| $f_{embed}$ | Embedding (settling) loss | 11 | µm | assumption: Three interfaces, machined. | 0 … 100 µm |
| $M_A$ | Tightening torque | 80 | N·m | spec: Assembly procedure AP-7 | 0.1 … 10000 N·m |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| K | Nut factor (nut-factor method); $F_M$ is the maximum preload, so enter the smallest value the joint can have | 0.2 | | assumption: Not used by the friction method. | 0.1 … 0.35 |
| $\mu_G$ | Thread friction coefficient; $F_M$ is the maximum preload, so enter the smallest value the joint can have | 0.12 | | assumption: Lightly oiled zinc-plated thread. | 0.06 … 0.25 |
| $\mu_K$ | Head / nut bearing friction coefficient; $F_M$ is the maximum preload, so enter the smallest value the joint can have | 0.12 | | assumption: Lightly oiled head bearing. | 0.06 … 0.25 |
| $\nu_{perm}$ | Permitted assembly utilisation ν: the equivalent stress at assembly over the proof strength (VDI 2230 R7; 0.9 is the usual target, 1.0 is the limit at which the bolt yields in tightening) | 1 | | default | 0.5 … 1 |
| $\alpha_A$ | Tightening scatter factor ($F_{M,max}$ / $F_{M,min}$) | 1.6 | | assumption: Torque wrench, VDI 2230 Table A8 order. | 1 … 4 |
| $F_A$ | Axial working load (per bolt) | 60 | kN | spec: LOAD-CASE-3 | 0 … 5000 kN |
| $F_{KR,req}$ | Required residual clamp force | 5 | kN | spec: SPEC-FLG §6 sealing | 0 … 5000 kN |
| $\tau_B$ | Nut / internal-thread material shear strength | 400 | MPa | assumption: Nut material shear strength. | 50 … 1200 MPa |
| $m_{eff}$ | Thread engagement length provided | 12 | mm | drawing: DRG-FLG-02 rev A | 1 … 200 mm |
| $S_{strip}$ | Safety factor on stripping | 1.25 | | assumption: Stripping safety factor. | 1 … 3 |

## 3. Assumptions

A1. The joint is concentrically clamped and loaded, and the members behave as two 30° frusta in series. (Eccentric clamping and loading (VDI 2230 full procedure) are outside this sheet.)
A2. Thread torsion is taken at 50 % in the equivalent stress for the working state. (Common VDI 2230 practice ($k_\tau$ = 0.5).)
A3. The thread torsion modulus is taken at the minor diameter $d_3$, which is conservative. (With the VDI 2230 mean diameter the torsion stress is about 14 % lower and $\nu_M$ about 5 % lower at the defaults.)
A4. $F_M$ is the maximum preload, from the smallest friction and nut factor, and $F_{M,min}$ is the minimum after scatter. (VDI 2230 R2: the maximum preload comes from the minimum friction coefficients.)

A5. The working load pulsates from 0 to $F_A$, so the fatigue amplitude is $\Phi F_A / (2 A_s)$. (Compare it with the bolt's endurance limit as entered.)
A6. Thread stripping uses a shear area of half the pitch per unit length on the nominal diameter. (A conservative simplification of ISO 898-2 / VDI 2230 §5.5.5.)
A7. $\sigma_{a,limit}$ (Bolt endurance limit (stress amplitude)) is taken as 50 MPa. (Rolled-before-heat-treatment M12, VDI 2230 order of magnitude.)
A8. $n_{load}$ (Load introduction factor (0 = at the clamped faces, 1 = under the head)) is taken as 0.5. (Load introduced at mid-thickness.)
A9. $f_{embed}$ (Embedding (settling) loss) is taken as 11 µm. (Three interfaces, machined.)
A10. K (Nut factor (nut-factor method); $F_M$ is the maximum preload, so enter the smallest value the joint can have) is taken as 0.2. (Not used by the friction method.)
A11. $\mu_G$ (Thread friction coefficient; $F_M$ is the maximum preload, so enter the smallest value the joint can have) is taken as 0.12. (Lightly oiled zinc-plated thread.)
A12. $\mu_K$ (Head / nut bearing friction coefficient; $F_M$ is the maximum preload, so enter the smallest value the joint can have) is taken as 0.12. (Lightly oiled head bearing.)
A13. $\alpha_A$ (Tightening scatter factor ($F_{M,max}$ / $F_{M,min}$)) is taken as 1.6. (Torque wrench, VDI 2230 Table A8 order.)
A14. $\tau_B$ (Nut / internal-thread material shear strength) is taken as 400 MPa. (Nut material shear strength.)
A15. $S_{strip}$ (Safety factor on stripping) is taken as 1.25. (Stripping safety factor.)

## 4. Working

[24] $d_2 = d - 0.6495 * p_{thread}$ [2](§9.1.6)
$= 12\ mm - 0.6495 \times 1.75\ mm$
$= 10.863\ mm$
Pitch diameter

[25] $d_3 = d - 1.2269 * p_{thread}$ [2](§9.1.6)
$= 12\ mm - 1.2269 \times 1.75\ mm$
$= 9.8529\ mm$
Minor diameter

[26] $A_s = pi / 4 * ((d_2 + d_3) / 2)^2$ [2](§9.1.6)
$= \pi / 4 \times ((10.863\ mm + 9.8529\ mm) / 2)^2$
$= 84.266\ mm^2$
Tensile stress area

[27] $A_N = pi / 4 * d^2$
$= \pi / 4 \times (12\ mm)^2$
$= 113.1\ mm^2$
Nominal shank area

[28] $A_{d3} = pi / 4 * d_3^2$
$= \pi / 4 \times (9.8529\ mm)^2$
$= 76.247\ mm^2$
Minor-diameter area

[29] $D_{Km} = (d_w + d_h) / 2$
$= (18\ mm + 13\ mm) / 2$
$= 15.5\ mm$
Mean bearing diameter under the head

[30] $F_M = M_A / (0.16 * p_{thread} + 0.58 * d_2 * \mu_G + 0.5 * D_{Km} * \mu_K)$ [1](§5.4.3)
$= 80\ N·m / (0.16 \times 1.75\ mm + 0.58 \times 10.863\ mm \times 0.12 + 0.5 \times 15.5\ mm \times 0.12)$
$= 40.69\ kN$
Assembly preload (friction)

[31] $F_{M,min} = F_M / \alpha_A$ [1](R2)
$= 40.69\ kN / 1.6$
$= 25.431\ kN$
Minimum preload after scatter

[32] $\sigma_M = F_M / A_s$
$= 40.69\ kN / 84.266\ mm^2$
$= 482.87\ MPa$
Preload stress
[33] $M_G = F_M * (0.16 * p_{thread} + 0.58 * d_2 * \mu_G)$ [1](§5.4.3)
$= 40.69\ kN \times (0.16 \times 1.75\ mm + 0.58 \times 10.863\ mm \times 0.12)$
$= 42.158\ N \cdot m$
Thread torque under preload
[34] $W_p = pi / 16 * d_3^3$
$= \pi / 16 \times (9.8529\ mm)^3$
$= 187.81\ mm^3$
Polar section modulus at the minor diameter
[35] $\tau_M = M_G / W_p$
$= 42.158\ N \cdot m / 187.81\ mm^3$
$= 224.47\ MPa$
Thread torsional stress under preload
[36] $\sigma_{red,M} = sqrt(\sigma_M^2 + 3 * \tau_M^2)$ [1](R8)
$= sqrt((482.87\ MPa)^2 + 3 \times (224.47\ MPa)^2)$
$= 619.94\ MPa$
Equivalent stress at assembly (full torsion)
[37] $v_M = \sigma_{red,M} / R_{p02}$
$= 619.94\ MPa / 640\ MPa$
$= 0.96866$
Utilisation of proof strength at assembly
[38] $\sigma_{perm} = v_{perm} * R_{p02}$ [1](R7)
$= 1 \times 640\ MPa$
$= 640\ MPa$
Permitted equivalent stress at assembly (permitted v times the proof strength)
[39] $u_{assembly} = v_M / v_{perm}$
$= 0.96866 / 1$
$= 0.96866$
Assembly utilisation over the permitted v
[40] $\delta_S = 0.4 * d / (E_S * A_N) + l_k / (E_S * A_s) + 0.5 * d / (E_S * A_{d3}) + 0.4 * d / (E_S * A_N)$ [1](§5.1.1)
$= 0.4 \times 12\ mm / (205\ GPa \times 113.1\ mm^2) + 40\ mm / (205\ GPa \times 84.266\ mm^2) + 0.5 \times 12\ mm / (205\ GPa \times 76.247\ mm^2) + 0.4 \times 12\ mm / (205\ GPa \times 113.1\ mm^2)$
$= 0.0000031135\ m/kN$
Bolt resilience (head, shank, engaged thread, nut)
[41] $k_S = 1 / \delta_S$
$= 1 / 0.0000031135\ m/kN$
$= 321190\ kN/m$
Bolt stiffness
[42] $t_f = l_k / 2$
$= 40\ mm / 2$
$= 20\ mm$
Frustum height (two equal frusta)
[43] $k_f = 0.5774 * pi * E_P * d / ln(((1.155 * t_f + d_w - d_h) * (d_w + d_h)) / ((1.155 * t_f + d_w + d_h) * (d_w - d_h)))$ [3](eq. 8-20)
$= 0.5774 \times \pi \times 205\ GPa \times 12\ mm / ln((1.155 \times 20\ mm + 18\ mm - 13\ mm) \times (18\ mm + 13\ mm) / ((1.155 \times 20\ mm + 18\ mm + 13\ mm) \times (18\ mm - 13\ mm)))$
$= 3815600\ kN/m$
Stiffness of one frustum

[44] $k_P = k_f / 2$ [3](eq. 8-21)
$= 3815600\ kN/m / 2$
$= 1907800\ kN/m$
Clamped members' stiffness (two frusta in series)
[45] $\Phi_K = n_{load} * k_S / (k_S + k_P)$ [1](R3)
$= 0.5 \times 321190\ kN/m / (321190\ kN/m + 1907800\ kN/m)$
$= 0.072047$
Load factor
[46] $F_Z = f_{embed} * (k_S * k_P) / (k_S + k_P)$ [1](R4)
$= 11\ \mu m \times 321190\ kN/m \times 1907800\ kN/m / (321190\ kN/m + 1907800\ kN/m)$
$= 3.024\ kN$
Preload loss from embedding
[47] $F_{SA} = \Phi_K * F_A$
$= 0.072047 \times 60\ kN$
$= 4.3228\ kN$
Additional bolt load from the working load
[48] $F_{PA} = (1 - \Phi_K) * F_A$
$= (1 - 0.072047) \times 60\ kN$
$= 55.677\ kN$
Load relieved from the members
[49] $F_{KR} = \max(F_{M,min} - F_{PA} - F_Z, 0\ kN)$ [1](R7)
$= \max(25.431\ kN - 55.677\ kN - 3.024\ kN, 0\ kN)$
$= 0\ kN$
Residual clamp force (minimum; zero once the joint opens)
[50] $F_{S,max} = \max(F_M + F_{SA}, F_A)$
$= \max(40.69\ kN + 4.3228\ kN, 60\ kN)$
$= 60\ kN$
Maximum bolt load (maximum preload + additional load; the whole working load once the joint opens)
[51] $\sigma_{z,max} = F_{S,max} / A_s$
$= 60\ kN / 84.266\ mm^2$
$= 712.03\ MPa$
Maximum bolt tensile stress
[52] $\sigma_{red,B} = \mathrm{sqrt}(\sigma_{z,max}\text{^}2 + 3 * (0.5 * \tau_M)\text{^}2)$ [1](R8)
$= \mathrm{sqrt}((712.03\ MPa)\text{^}2 + 3 \times (0.5 \times 224.47\ MPa)\text{^}2)$
$= 738.09\ MPa$
Equivalent stress in the working state (50 % torsion)
[53] $u_{yield} = \sigma_{red,B} / R_{p02}$
$= 738.09\ MPa / 640\ MPa$
$= 1.1533$
Yield utilisation in the working state
[54] $\sigma_a = (F_{S,max} - F_M) / (2 * A_s)$ [1](R9)
$= (60\ kN - 40.69\ kN) / (2 \times 84.266\ mm^2)$
$= 114.58\ MPa$
Fatigue stress amplitude (pulsating 0…$F_A$ load)
[55] $u_{fatigue} = \sigma_a / \sigma_{a,limit}$
$= 114.58\ MPa / 50\ MPa$
$= 2.2916$
Fatigue utilisation
[56] $m_{min} = S_{strip} * F_{S,max} / (0.5 * pi * d * \tau_B)$ [1](§5.5.5)
$= 1.25 \times 60\ kN / (0.5 \times \pi \times 12\ mm \times 400\ MPa)$
$= 9.9472\ mm$
Thread engagement needed against stripping

[57] $u_{strip} = m_{min} / m_{eff}$
$= 9.9472\ mm / 12\ mm$
$= 0.82893$
Engagement utilisation
[58] $F_{A,sep} = (F_{M,min} - F_Z) / (1 - \Phi_K)$ [1](R7)
$= (25.431\ kN - 3.024\ kN) / (1 - 0.072047)$
$= 24.147\ kN$
Working load at which the members separate (minimum preload, after embedding)
[59] $F_{proof} = R_{p02} * A_s$
$= 640\ MPa \times 84.266\ mm^2$
$= 53.93\ kN$
Bolt load at the proof (yield) strength
[60] $n_0 = F_{A,sep} / F_A$
$= 24.147\ kN / 60\ kN$
$= 0.40245$
Load factor against separation ($F_{A,sep} / F_A$)
[61] $F_{A,yield} = \min((F_{proof} - F_M) / \Phi_K, F_{proof})$
$= \min((53.93\ kN - 40.69\ kN) / 0.072047, 53.93\ kN)$
$= 53.93\ kN$
Working load at which the bolt reaches its proof load (separation included)
[62] $n_L = F_{A,yield} / F_A$
$= 53.93\ kN / 60\ kN$
$= 0.89884$
Load factor against axial yield of the bolt, torsion excluded

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $F_M$ | 40.69 kN | Assembly preload (maximum) |
| $F_{KR}$ | 0 kN | Residual clamp |
| $F_{S,max}$ | 60 kN | Maximum bolt load |
| $\Phi_K$ | 0.072047 | Load factor |
| $\sigma_{red,B}$ | 738.1 MPa | Working equivalent stress |
| $n_0$ | 0.40244 | Load factor against separation |
| $n_L$ | 0.89884 | Load factor against axial yield |
| $A_s$ | 84.27 mm² | Tensile stress area |
| $F_{M,min}$ | 25.43 kN | Minimum preload |
| $\sigma_M$ | 482.9 MPa | Preload stress |
| $k_S$ | 321186 kN/m | Bolt stiffness |
| $k_P$ | 1907819 kN/m | Members' stiffness |
| $F_Z$ | 3.024 kN | Embedding loss |
| $F_{SA}$ | 4.323 kN | Additional bolt load |
| $F_{A,sep}$ | 24.15 kN | Separation load |
| $\sigma_{z,max}$ | 712.0 MPa | Maximum bolt tensile stress |
| $\sigma_a$ | 114.6 MPa | Fatigue amplitude |
| $m_{min}$ | 9.947 mm | Engagement needed |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $\nu_M$ | 0.97 | Assembly utilisation of proof strength |
| $u_{assembly}$ | 0.97 | Assembly utilisation over the permitted ν |
| $u_{yield}$ | 1.16 | Yield utilisation in the working state |
| $u_{fatigue}$ | 2.30 | Fatigue utilisation |
| $u_{strip}$ | 0.83 | Engagement utilisation |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| FAIL | Residual clamp force against separation | 0 kN | > 0 kN | — | [1](R7) |
| FAIL | Required residual clamp force against the residual clamp force | 0 kN | ≥ 5 kN | — | [1](R7) |
| FAIL | Fatigue stress amplitude against the endurance limit | 114.6 MPa | 50 MPa | 2.30 | [1](R9) |
| FAIL | Working equivalent stress against the proof strength | 738.1 MPa | 640 MPa | 1.16 | [1](R8) |
| PASS | Assembly equivalent stress against the permitted stress | 619.9 MPa | 640 MPa | 0.97 | [1](R7) |
| PASS | Thread engagement needed against the engagement provided | 9.947 mm | 12 mm | 0.83 | [1](§5.5.5) |

## 7. Applicability

WARNING $M_A$: the preload alone takes the bolt above 90 % of its proof strength. [preload-yield]
WARNING $F_{KR}$: the residual clamp force is not positive. The joint separates under the working load, and the bolt carries all of it. [separation]

Defaults used (inputs left empty):
$\nu_{perm}$: not entered, so the default of 1 is used.

## 8. Verification

```
Template mech.bolted-joint@1.3.1 Content hash c143be03cdb54ee7 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

## 9. References

1. Verein Deutscher Ingenieure. Systematic Calculation of Highly Stressed Bolted Joints: Joints With One Cylindrical Bolt. VDI 2230 Part 1:2015. Beuth Verlag; 2015.
2. International Organization for Standardization. Mechanical Properties of Fasteners Made of Carbon Steel and Alloy Steel—Part 1: Bolts, Screws and Studs With Specified Property Classes—Coarse Thread and Fine Pitch Thread. ISO 898-1:2013. ISO; 2013.
3. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.